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SAMPLE

Digital SAT Reference Test

March 2024 *Question Booklet*

Test date: ..... / ..... / .....

CANDIDATE NAME			
CENTRE NUMBER		CANDIDATE NUMBER	

Time allowed: 02 hours 14 minutes

Reading and Writing Section: 64 minutes

Math Section: 70 minutes

You must answer on the **question booklet**.

INSTRUCTIONS

- **Reading and Writing Section:** answer **ALL** questions.
- **Math Section:** answer **ALL** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Read the instructions for each part of the paper carefully.
- Write your answer to each question in the space provided.
- Do not use an erasable pen or correction fluid.
- You may use a calculator on the Math section.
- You must complete the question paper within the time limit.
- At the end of the test, hand in this question booklet to the supervisor.

INFORMATION

- This question paper consists of **44** pages.

  *Reading and Writing Section: Module 1 pp. **2-11**; Module 2 pp. **12-21***

  *Math Section: Module 1 pp. **22-25**; Module 2 pp. **26-29***
- Each section has **TWO** modules.
- The total questions for this paper are **98**.

  *Reading and Writing Section: **54** questions (**27** questions for each module)*

  *Math Section: **44** questions (**22** questions for each module)*

- DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO –

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Reading and Writing

– 27 Questions | 32 minutes –

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The following text is from Rudolfo Anaya's 1972 novel *Bless Me, Utma*. The young narrator is eating breakfast with his two sisters in their home in rural New Mexico.

I sat across the table from Deborah and Teresa and ate my atole and the hot tortilla with butter. I said very little. I usually spoke very little to my two sisters. They were older than I and they were very close. They usually spent the entire day in the attic, playing dolls and giggling. I did not ***concern myself with those*** *things.*

As used in the text, what does the phrase "***concern myself with***" most nearly mean?

A craft

B get angry about

C care about

D earn

2

The percentage of US forest land that a 2023 federal report identified as being either mature or old growth exceeds other recent estimates. Given how little _____ there is among scientists regarding the scope of these categories, this discrepancy shouldn't be surprising: forest researchers regularly dispute one another's classifications.

Which choice completes the text with the most logical and precise word or phrase?

A interest

B deliberation

C vigilance

D consensus

3

Karam Kang has demonstrated that lobbying does little to alter probability that a particular energy policy under consideration by the United States Congress will be enacted, but lobbying is not as _____ finding seems to suggest. As Kang herself notes, lobbying can shape which policy proposals members of Congress bring forward for consideration in the first place.

Which choice completes the text with the most logical and precise word or phrase?

A counterproductive

B cynical

C biased

D ineffectual

4

Though it does not guarantee a book's commercial success, _____ can play a big role in that success – a well-executed marketing campaign helped to make Danielle Steel's novel The Gift, the fourth-best-selling novel of 1994.

Which choice completes the text with the most logical and precise word or phrase?

A originality

B publicity

C complexity

D practicality

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5

People sometimes dismiss a claim if it comes from a source they regard as self-interested, but from a strictly logical perspective, the source of a claim is _____: it has no direct bearing on whether the claim is true.

Which choice completes the text with the most logical and precise word or phrase?

A implicit

B indisputable

C indistinct

D irrelevant

6

When people think of dinosaurs with feathers, they typically think of winged dinosaurs, such as the four-winged Microraptor. ***However, many dinosaurs that didn't have wings also had feathers on their bodies.*** *For instance, research indicates that the wingless, speedy Ornithomimus likely had feathers.*

Which choice best describes the function of the ***underlined*** sentence in the text as a whole?

A It reveals that the dinosaurs mentioned earlier in the text used their feathers for an important purpose.

B It gives an example of a dinosaur with the characteristics mentioned earlier in the text.

C It defines a term related to the type of dinosaur feathers mentioned earlier in the text.

D It indicates that the dinosaurs mentioned earlier in the text weren't the only ones that had feathers.

7

The following text is from Annie Dillard's 1987 autobiographical novel *An American Childhood*. The narrator is a young girl living in Pittsburgh.

I walked. My mother had given me the freedom of the streets as soon as I could say our telephone number. I walked and memorized the neighborhood. I made a mental map and located myself upon it. At night in bed I rehearsed the small world's scheme and set challenges: ***Find the store using backyards only. Imagine a route from the school to my friend's house.***

Which choice best describes the function of the ***underlined*** portion in the text as a whole?

A It describes the narrator trying to memorize her telephone number.

B It provides examples of what the narrator thinks about at night.

C It gives directions to the narrator's favorite local store.

D It portrays the narrator's relationship with her mother.

8

Gus Edwards's atmospheric play Black Body Blues was produced in 1978 by the groundbreaking Negro Ensemble Company (NEC). NEC cofounder Douglas Turner Ward, who worked as an actor, director, and playwright, had met actor and producer Robert Hooks while they were performing in a 1960 touring production of Lorraine Hansberry's play A Raisin in the Sun. Together, they envisioned a theater company that would nurture and showcase the work of Black theater professionals. Since NEC's founding in 1967, its workshops and performances have given Black playwrights, including Edwards, a forum for their compelling stories.

Which choice best states the main purpose of the text?

A To show how the play *A Raisin in the Sun* influenced Ward and Hooks.

B To bring attention to a work by Gus Edwards.

C To describe how Ward and Hooks's creative partnership changed over time.

D To discuss the origin and importance of the *NEC*.

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9

The food industry has long used thermal technologies to preserve food in large batches. Recent advancements in infrared heating were made through research in Mexico on the preservation of bean flour. Infrared heating is generally considered to be an improvement over more conventional thermal preservation methods: whereas conventional methods transfer energy from the surface of a food to its interior, infrared heating uses high-intensity infrared light to generate heat within the food itself, thus reducing industrial cooking times.

Based on the text, what is one disadvantage of some conventional thermal methods?

A They are especially vulnerable to technological breakdowns.

B They are only effective in heating foods that are lightweight and small in size.

C Their cooking times are longer than those of other methods.

D They require more energy than other methods do.

10

"Cocoa" is an example of a loanword – that is, a word that originated in one language and was later adopted by another. The word came to English indirectly from cacao, the Spanish word for the plant that chocolate is made from. Spanish had borrowed it from Nahuatl, an Indigenous language of Central Mexico, in which the word's original form is cacahuatl. "Iguana" is also Indigenous in origin and entered English through Spanish. But in this case, the original source was Taino, a language of the Caribbean islands, in which the word for the group of related lizard species is twana.

The author makes which point about the Spanish language?

A It has served as a medium through which Indigenous languages have influenced English.

B Its contribution to English vocabulary roughly equals the collective contribution by Indigenous languages.

C It has borrowed words from Indigenous languages and contributed words to them.

D It adopted Nahuatl and Taino words in approximately equal numbers.

11

Total Areas of hive Tribal Nations around the United States

Tribal Nation	Location	Area *(square miles)*
Cherokee Nation	Oklahoma	6,963
Northern Cheyenne Tribe	Montana	707
Mandan, Hidatsa, and Arikara Nation	North Dakota	1,583
Colville Tribes	Washington	2,185
Muscogee Nation	Oklahoma	4,867

In terms of total area, the Navajo Nation is the largest tribal nation in the United States. It covers 24,425 square miles in Arizona, Colorado, and New Mexico. Most tribal nations in the US are much smaller. For example, the total area of the Muscogee Nation in Oklahoma is 4,867 square miles. And in Washington, the total area of the Colville Tribes is _____.

Which choice most effectively uses data from the table to complete the statement?

A 2,185 square miles

B 707 square miles

C 6,963 square miles

D 1,583 square miles

12

Millions of Metric Tons of Copper Mined in 1995 and 2020

Country	1995	2020
Poland	0.38	0.39
Kazakhstan	0.26	0.55
Chile	2.49	5.73
Canada	0.73	0.59

While doing research for a paper about copper mining, a student finds a table with information about four different countries. The student notes that the country that mined 0.55 million metric tons of copper in 2020 had mined _____.

Which choice most effectively uses data from the table to complete the statement?

A 0.26 million metric tons of copper in 1995

B 0.59 million metric tons of copper in 1995

C 2.49 million metric tons of copper in 1995

D 0.38 million metric tons of copper in 1995

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13

"A Pair of Silk Stockings" is an 1897 short story written by Kate Chopin. In the story, Chopin suggests that the protagonist, Mrs. Sommers, experiences a shift in her sense of self as a result of putting on certain articles of clothing, writing, _____.

Which quotation from *"A Pair of Silk Stockings"* most effectively illustrates the claim?

A "Mrs. Sommers was one who knew the value of bargains; who could stand for hours making her way inch by inch toward the desired object that was selling below cost."

B "She held back her skirts and turned her feet one way and her head another way as she glanced down at the polished pointed-tipped boots."

C "She would buy so and so many yards of percale for new shirt waists for the boys and Janie and Mag."

D "Her stockings and boots and well-fitting gloves had worked marvels in her bearing – had given her a feeling of assurance, a sense of belonging to the well-dressed multitude."

14

Average Hours Worked per Person per Year in 1950 and 2017

Country	1995	2017	Change in hours	Percent change in hours
Argentina	2,034	1,692	- 342	- 17%
Ireland	2,419	1,746	- 674	- 28%
Mexico	2,432	2,255	- 177	- 7%
United Kingdom	2,184	1,670	- 514	- 24%

(Calculations may be inexact due to rounding)

A student in an economics course is examining the decline since 1950 in average hours worked per person per year in various nations due to both increased productivity and the adoption of policies that limit working hours. The first task in this investigation is to determine how the decline in Argentina compares to that in other countries. The student finds that _____.

Which choice most effectively uses data from the table to complete the student's conclusion?

A the decline in number of hours worked was greater in Argentina than it was in Ireland, the United Kingdom, or Mexico

B while the number of hours worked rose in Argentina between 1950 and 2017, it declined in Ireland, the United Kingdom, and Mexico

C though the percent decrease in hours worked in Argentina was greater than that in Ireland and Mexico, it was less than that in the United Kingdom

D though the percent decrease in hours worked in Argentina was less than that in Ireland and the United Kingdom, it was greater than that in Mexico

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15

The Mammillaria cactus M. guerreronis occurs naturally only in the state of Guerrero in Mexico, and the smallness of its range makes it especially vulnerable to extinction. The traditional single-species approach to conservation emphasizes the need to focus on individual species most at risk, like M. guerreronis, but recently, conservationists have argued that an ecosystem-based approach that incorporates the many interactions between the climate, terrain, and various species of a given geographical area may lead to better outcomes for all the species in a given location. If this view is correct, the single-species approach to the conservation of M. guerreronis could thus _____.

Which choice most logically completes the text?

A lead to a better understanding of how the distribution of *Mammillaria* species throughout Mexico has affected their survival

B allow conservationists to better consider how climatic changes affecting *Guerrero* may reduce the number of species competing with *M. guerreronis*

C erroneously shift the focus of conservation efforts away from *M. guerreronis* itself

D fail to consider the ways in which the survival of *M. guerreronis* may be influenced by changes in the populations of other species that inhabit Guerrero

16

It is widely known that New York City's Museum of Modern Art (MoMA) _____ a vast collection of oil paintings, including Paul Klee's In the Grass and Fernand Léger's Contrast of Forms.

Which choice completes the text so that it conforms to the conventions of Standard English?

A houses

B to have housed

C housing

D to house

17

An accomplished birder might be able to spot thousands of different species over the course of a career, from Amadina erythrocephala to Arborophila rufogularis, but no birder has ever achieved the ultimate _____ an in-person glimpse of all 10,000-plus species of birds.

Which choice completes the text so that it conforms to the conventions of Standard English?

A goal catching

B goal. Catching

C goal: catching

D goal; catching

18

Today, the Michelin Guide is widely known as the arbiter of fine dining with its coveted 3-star rating being awarded to top restaurants like Kitcho Arashiyama in _____ when brothers Andre and Edouard Michelin created the guide in 1900, it was little more than a marketing gimmick – a way to increase profits for their tire company by encouraging their customers to drive across France, visiting the guide's recommended hotels and restaurants along the way.

Which choice completes the text so that it conforms to the conventions of Standard English?

A Kyoto; however,

B Kyoto,

C Kyoto, however,

D Kyoto

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19

When attempting to determine a fault's seismic history, geophysicists like Dr. Estella Atekwana at the University of Delaware rely in part on data about the fault's physical dimensions and geological _____ the Imperial fault in Imperial County, California, the US Geological Survey compiles such data in several ways: by exchanging information with state geological surveys, by linking maps of the fault with sources that offer detailed descriptions of it, and by maintaining a database of quantitative data such as the fault's length (46 km).

Which choice completes the text so that it conforms to the conventions of Standard English?

A features for

B features, for

C features. For

D features and for

20

In 1987, chemist Ahmed Zewail devised a novel technique in optical spectroscopy that allowed him to study ultrafast chemical reactions occurring within molecules. Years later, he _____ for this groundbreaking research with the Robert A. Welch Award and the Priestley Medal from the American Chemical Society and hence would become known as the father of femtochemistry.

Which choice completes the text so that it conforms to the conventions of Standard English?

A had been honored

B would be honored

C is honored

D will be honored

21

For Canada, a member of the North Atlantic Treaty Organization (NATO) since 1949, NATO's principle of collective defense confers both benefits and _____ organization's many members, nations as disparate as the US and Slovenia, are all bound to defend Canada, the reverse is also true.

Which choice completes the text so that it conforms to the conventions of Standard English?

A obligations: while the

B obligations; the

C obligations, while the

D obligations. The

22

If you could travel in a spaceship to Elara, one of Jupiter's many moons, you'd find a moon with a prograde orbit. This means that Elara orbits Jupiter in the same direction that Jupiter rotates on its axis. _____ at Pasiphae, another Jovian moon, you'd find an example of a retrograde orbit, with the moon revolving around Jupiter in the opposite direction.

Which choice completes the text so that it conforms to the conventions of Standard English?

A For example,

B Specifically,

C In other words,

D Elsewhere,

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[Page 7 - NOT FOR SALE]

19

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Which choice completes the text with the most logical transition?

A For example,

B Specifically,

C In other words,

D Elsewhere

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23

While researching a topic, a student has taken the following notes:

- *Scientists have long sought to determine the origin of glass in Chile's Atacama Desert.*
- *A 2017 study concluded that ancient grass fires had melted the area's sandy soil into glass.*
- *In 2021, a different study revealed that the mineral signatures of glass samples were consistent with the mineral signatures of comet samples collected by NASA.*
- *That study concluded that the glass had formed as a result of a cometary explosion close to the desert's surface.*

The student wants to describe how scientific understanding about the glass's origin has evolved. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A Mineral signatures of glass samples are consistent with those of comet samples collected by *NASA*, according to new research.

B A 2017 study concluded that ancient grass fires had caused the glass's formation, but new research suggests that the glass formed as a result of a cometary explosion close to the desert's surface.

C Scientists have long sought to determine the origin of the glass, with one study concluding that it formed when ancient grass fires melted the area's sandy soil.

D Other answer

24

While researching a topic, a student has taken the following notes:

- *Maya Lin is a US artist known for her memorials and large-scale installation artworks.*
- *She became famous in 1982 when she completed the Vietnam Veterans Memorial, which consists of two 246-foot granite walls.*
- *She completed Water Line in 2006.*
- *It is an installation composed of aluminum tubing that fills an entire gallery room.*
- *She completed Seven Earth Mountain in 2015.*
- *It is an installation composed of soil that fills an entire gallery room.*

Which choice most effectively uses information from the given sentences to emphasize a difference between *Water Line* and *Seven Earth Mountain*?

A Maya Lin's *Water Line* is composed of aluminum tubing; *Seven Earth Mountain*, by contrast, is composed of soil.

B Maya Lin is known for her memorials and installation art, such as *Water Line* and *Seven Earth Mountain*.

C After completing the *Vietnam Veterans Memorial*, Maya Lin completed *Water Line*, another large-scale work.

D The sprawling size of Maya Lin's *Vietnam Veterans Memorial* is echoed in *Water Line*, a work made of aluminum tubing that fills an entire gallery room.

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25

While researching a topic, a student has taken the following notes:

- *The Peloncillo-Pan Duro Mountains are a mountain range located in northwestern Mexico.*
- *The range is one of the dozens of "sky islands" in the southwestern US and northwestern Mexico.*
- *A sky island is an isolated mountain range whose environment differs drastically from that of the surrounding lowlands.*
- *The US Forest Service (USFS) said, "The mountains are 'islands' surrounded by deserts that are 'seas'."*
- *The USFS said, "Each Sky Island is a unique ecosystem."*

The student wants to explain what a sky island is. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A The Peloncillo-Pan Duro Mountains are an isolated mountain range located in northwestern Mexico whose environment differs drastically from that of the surrounding lowlands.

B A sky island is an isolated mountain range, such as the Peloncillo-Pan Duro Mountains in northwestern Mexico, whose environment differs drastically from that of the surrounding lowlands.

C The Peloncillo-Pan Duro Mountains, which are considered to be a sky island, are located in northwestern Mexico.

D The USFS considers the Peloncillo-Pan Duro Mountains to be a "unique ecosystem."

26

While researching a topic, a student has taken the following notes:

- *The Paralympic Games are a series of international sporting events involving athletes with an impairment.*
- *A gold medal indicates that an athlete finished in first place.*
- *Rolf Heinzmann competed as a Paralympic alpine skier from 1980 to 2002.*
- *He won fourteen medals, twelve of which were gold.*
- *Daniel Dias competed as a Paralympic swimmer from 2008 to 2020.*
- *He won thirty-eight medals, twenty-five of which were gold.*

Which choice most effectively uses information from the given sentences to emphasize the number of gold medals won by Daniel Dias?

A In his career as a Paralympic swimmer, Daniel Dias won thirty-eight total medals – twenty-five of them were gold.

B Both multiple gold medalists, Roll Heinzmann and Daniel Dias debuted at the Paralympic Games in 1980 and 2008, respectively.

C At the Paralympic Games, a gold medal indicates a first-place finish, and some accomplished athletes like Rolf Heinzmann have won many.

D An accomplished Paralympian with thirty-eight total medals, Daniel Dias competed as a swimmer from 2008 to 2020.

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27

While researching a topic, a student has taken the following notes:

- *The human tongue contains taste receptors for a rich, savory flavor called umami.*
- *Umami is triggered by the compounds in a variety of foods, including pork and tomato paste.*
- *Participants in a study tasted a sample of macrokelp, a type of brown seaweed.*
- *They rated its umami intensity as moderate.*
- *The participants tasted a sample of ma-konbu, another type of brown seaweed.*
- *They rated its umami intensity as high.*

The student wants to make a generalization about brown seaweed. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A Macrokelp is a type of brown seaweed, as is ma-konbu.

B In a research study, participants tasted samples of macrokelp and ma-konbu, two types of brown seaweed.

C Participants in a study rated the umami intensity of various seaweeds.

D Some types of brown seaweed trigger umami flavor in human taste buds.

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STOP

If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.

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Reading and Writing

– 27 Questions | 32 minutes –

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

Although Oscar Micheaux's 1920 film Within Our Gates and Ossie Davis's 1970 film Cotton Comes to Harlem may have had some _____ at the time of their initial release, their place in critics' estimations is now more secure. In 2018, for example, critics for the New York Times described the former as "stunning" and the latter as "especially pointed."

Which choice completes the text with the most logical and precise word or phrase?

A collaborators

B imitators

C precursors

D detractors

2

Political scientists have found that although voters claim to prefer candidates who have nuanced perspectives on issues and who show a willingness to compromise, when asked to compare speeches expressing such views with speeches expressing _____ views, voters tend to regard the unyielding rhetoric of the latter more favorably.

Which choice completes the text with the most logical and precise word or phrase?

A dogmatic

B vacillating

C banal

D disingenuous

3

A number of recording artists _____ distinctions between Indigenous music and other musical styles. For example, Choctaw singer-songwriter Samantha Crain incorporated elements of Choctaw hymns into folk-pop on her single "When We Remain," and Oji-Cree/Mi'kmaq musician Anachnid combined Indigenous flute and powwow vocals with electronic music on her album Dreamweaver.

Which choice completes the text with the most logical and precise word or phrase?

A disguise

B reject

C observe

D replace

4

Today composer Scott Joplin is mainly celebrated for his catchy ragtime pieces "Maple Leaf Rag" and "The Entertainer." However, by overlooking his less famous works, listeners will miss the full range of Joplin's creativity. For instance, his waltz "Pleasant Moments" and his opera "Treemonisha" skillfully blend ragtime and classical music. These masterpieces deserve as much fame as Joplin's biggest hits.

Which choice best states the main purpose of the text?

A To argue that more attention should be given to Joplin's lesser-known works.

B To describe the similarities and differences between ragtime music and opera.

C To explain how Joplin learned to compose and perform ragtime music.

D To encourage music lovers to listen to music by many different composers.

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5

Desolation Island, first published in 1978, is a novel in Patrick O'Brian's Aubrey/Maturin series, which includes twenty completed books. Some critics have found fault with the abrupt endings of Desolation Island and other books in the series, saying that they do not finish conclusively but arbitrarily stop. But other critics argue that the books should not be thought of as discrete texts with traditional beginnings and endings but as a single incredibly long work, similar to other multivolume stories, such as John Galsworthy's The Forsyte Saga.

Which choice best states the main purpose of the text?

A To explain why many critics find the Aubrey/Maturin novels to be remarkably entertaining despite flaws in the novels' structures.

B To describe a characteristic of the Aubrey/Maturin novels and offer two differing viewpoints on this characteristic.

C To argue that the Aubrey/Maturin series should have the literary renown of *The Forsyte saga*.

D To present a reason the unusual structure that O'Brian uses for *Desolation Island* makes it one of his most complex books.

6

Text 1

Isaac Asimov, author of The Naked Sun and Pebble in the sky, is highly regarded despite his mediocre writing style. His prose is workmanlike; his characters are flat and discuss ideas rather than emotions. *That his work is enjoyable despite this is a testament to his prodigious imagination – even if people read his books only for the ideas, they will have plenty to consider.*

Text 2

Asimov is critiqued for his style, but it is wrong to fault a writer for failing to do what he never intended to do. For example, although most of his novel Foundation consists of people discussing science and politics and we find out little about his characters' lives, Asimov wanted to convey the vast sweep of human history over centuries, and one of his points is that at such a timescale, individuals don't matter. Thus, his lack of characterization is central to Asimov's thematic aims.

Based on the texts, how would the author of **Text 2** most likely respond to the evaluation of Asimov's writing style in the ***underlined*** claim in **Text 1**?

A By asserting that Asimov's minimal characterization is likely a conscious choice that helps him convey an important idea.

B By arguing that Asimov's writing style should be judged against the styles of historians rather than those of fiction writers.

C By agreeing that Asimov's prose is flawed but disagreeing about the quality of his characters.

D By pointing out that the flatness of Asimov's characters is a feature of *The Naked Sun* and *Pebble in the Sky* but not of *Foundation*.

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7

Reyjane P. Oliveira and colleagues relied on historical DNA (hiDNA) – genomic data incidentally preserved in specimens housed in natural history collections – to investigate the evolutionary trajectory of the Malagasy grass genus Chasechloa. Although this approach offers unique benefits, such as the ability to ascertain the biogeographical origins of organisms, it remains a relatively underutilized resource because hDNA is often to some extent degraded, a situation not easily remediable under current methodological paradigms and with extant DNA extraction and analysis technologies.

Information in the text best supports which statement about hDNA?

A It may yield insights that other types of genomic data cannot.

B It may be underused because of its controversial status among scientists.

C It has thus far proved valuable mainly to researchers studying flora.

D It tends to be much more degraded than other types of DNA of comparable age.

8

Meredith F. Protas and colleagues have explored how convergent evolution – a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages – can result from a genetic mechanism shared by both lineages. Meanwhile, Cynthia C. Steiner and colleagues have investigated how convergence occurs through different genetic mechanisms, but the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

Which choice best states the main idea of the text?

A Green and Extavour's study was conducted using data from the studies by Protas and colleagues and Steiner and colleagues.

B Green and Extavour's study addresses convergent evolution more comprehensively than the studies by Protas and colleagues and Steiner and colleagues do.

C Both the study by Protas and colleagues and that by Green and Extavour compare convergence through shared genetic mechanisms to convergence through different genetic mechanisms.

D Before Green and Extavour's study, convergent evolution was assumed to require a shared genetic mechanism between two lineages that share the trait.

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9

Over time, marine mollusks called nautilids continually add segments to their shells, and the chemical composition of each segment is influenced by the available materials in the surrounding water when the segment forms. Paleontologist Amane Tajika and colleagues analyzed the septa (the walls between shell chambers) of one male (samples labeled "M") and one female (samples labeled "F") nautilid, finding that sample M06 had the same concentration of oxygen-18 as did water approximately 120 meters deep and that the oxygen-18 concentration in sample F14 matched that of water at approximately 385 meters deep. Tajika et al. concluded that sample M06 formed at a higher point in the water column than did F14.

Which finding, if true, would most directly support the researchers' conclusion?

A In nautilids, sex predicts septa growth rate more strongly than does oxygen-18 concentration in the surrounding water.

B Regardless of sex, nautilids spend most of their lives at depths of 50 meters or greater, where oxygen 18 concentrations typically differ from those higher in the water column.

C Male nautilids tend to show higher concentrations of oxygen-18 than do female nautilids inhabiting the same point in the water column.

D Amone nautilids, there is no effect of sex on the uptake of oxygen-18 from water or its deposition in septa.

10

Callie W. Babbitt, Hema Madaka, and colleagues assembled a database of materials used in consumer electronics by studying products in the lab and by gathering data from similar product studies. The team gave each of these studies a rating for level of traceability (with a higher rating for clearer descriptions of procedures) and for category consistency (with a higher rating for using materials categories more closely aligned with the categories in the team's database). Based on these ratings, a second research team concluded that the methodology was better explained in a study by Greg L. Kozak and Gregory A. Keoleian than it was in a study by Martin Streicher-Porte and colleagues.

Which finding, if true, would most directly challenge the second research team's conclusion?

A The study by Kozak and Keoleian had a lower consistency rating than the study by Streicher-Porte and colleagues did.

B The study by Kozak and Keoleian had a high consistency rating and a high traceability rating.

C The study by Streicher-Porte and colleagues had a medium consistency rating and a medium traceability rating.

D The study by Kozak and Keoleian had a lower traceability rating than the study by Streicher-Porte and colleagues did.

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11

Periods of subfreezing temperatures in Alaska have been growing shorter in response to climate warming, potentially enabling increased carbon dioxide (CO_2) absorption through greater productivity of creeping dogwood (Cornus canadensis) plants and other vegetation, but also potentially enabling Early snow melt reduces the amount of soil moisture available for the increased CO_2 output through greater heterotrophic respiration (CO_2 generated by the activity of soil microorganisms). Hydrologist Yonghong Yi and her colleagues developed a model incorporating numerous inputs – years of solar radiation and snow cover data among them – to evaluate the effects of warming on the CO_2 balance in Alaska, concluding that net CO_2 is likely to increase if warming hastens spring snow melt.

Which finding, if true, would most directly support the researchers' conclusion?

A Early snow melt reduces the amount of soil moisture available for the growth of plant species such as *C. canadensis*, and it raises the rate of heterotrophic respiration.

B Early snow melt extends the period in which plant species such as *C. canadensis* can absorb CO_2, but has no effect on the rate of heterotrophic respiration.

C Early snow melt slows the growth of plant species such as *C. canadensis* by reducing soil insulation, and it suppresses heterotrophic respiration.

D Early snow melt amplifies the effect of solar radiation on the growth of plant species such as *C. canadensis* and on the rate of heterotrophic respiration.

12

In Armenia, use of solid fuel (e.g., coal, wood) as a share of total household about household fuel use fell by approximately 90 percent between 2000 and 2018; such shifts are typically explained by appeal to the energy ladder, a model holding that fuel choice is mediated mainly by household income (specifically, high-technology fuels displace solid fuels as incomes rise). Boukary Ouedraogo's study of fuel use in Burkina Faso shows how reductive this model is, however: although income of course constrained fuel choice, several factors, including the frequency with which certain dishes are cooked, influenced decisions.

Based on the text, the author would most likely agree with which statement about household income?

A It constrains the amount of fuel households use but not the type of fuel they use.

B It affects household fuel choice but not for the reasons assumed by the energy ladder model.

C It can explain some but not all of the differences in fuel choice across households.

D It is often said to influence household fuel choice but actually does not.

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13

The Socio-Economic Panel (SOEP) is a longitudinal study surveying approximately 12,000 individuals in Germany to glean extended trends in childhood development. To carry out the necessary sampling of the population over many years, SORP needs extensive financial support, but this method provides valuable insights into causal relationships. However, when questions of causation are irrelevant, as with a transportation study seeking only to reveal the number of people per day who use a city's public transportation system, longitudinal methods are unnecessary, and so _____.

Which choice most logically completes the text?

A the expense of *SOEP* is likely greater than the cost of longitudinal studies of transportation

B conclusions drawn from *SOEP* are likely to be more authoritative than those from the transportation study

C longitudinal methods are suitable for studies of childhood development but ought to be avoided for those of transportation

D the success of the transportation study likely requires significantly less financial outlay than that needed for *SOEP*

14

The many editions of James Joyce's 1922 novel Ulysses are not textually identical, and scholars debate which versions reflect Joyce's authorial intent. One no longer widely read edition is the 1984 "critical and synoptics edition" edited by Hans Walter Gabler, which followed French and German editorial theories rather than editorial traditions of the United States and (United Kingdom and which was later found to have introduced errors due to Gabler's choice to consult facsimile manuscripts rather than using only originals. However, few Joyce scholars worldwide had expertise in such textual issues, and most of those who did worked on the edition with Gabler. So, it is unsurprising that initial scholarly reviews of the 1984 edition were mostly _____.

Which choice most logically completes the text?

A negative, since any scholar with expertise in editorial theories of the United States and United Kingdom as well as French and German editorial theories most likely worked with Gabler on the 1984 edition and would therefore not review it

B negative, since those Joyce scholars with the necessary expertise to write a review of the 1984 edition would be aware that facsimile manuscripts cannot be produced with a high enough fidelity to the original to ensure that relying on them will not introduce editorial errors

C positive, since scholars who reviewed the 1984 edition were unaffiliated with its production and were mostly either Joyce specialists who were largely unfamiliar with editorial theories and practices or specialists in such theories and practices who were insufficiently familiar with Joyce

D Other answer

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15

Arthurian legends (tales related to the character of King Arthur) derive from many sources, such as Annales Cambriae, composed around 970, and the Mabinogion from the 12th and 13th centuries. One of the most significant sources, Geoffrey of Monmouth's History of the Kings of Britain, was written in Latin in the 1130s; some material from it was later adapted by the Norman poet Wace into the Roman de Brut in 1155. But while no source before 1155 includes references to the famous Round Table at which Arthur's knights assembled, both the Roman de Brut and Sir Thomas Malory's 15th century compilation of Arthurian legends, Le Morte d'Arthur do. It can therefore be inferred that _____.

Which choice most logically completes the text?

A *Le Morte d'Arthur* is more historically accurate than *History*, because the Mabinogion had not been written when Geoffrey of Monmouth was writing his work

B Geoffrey of Monmouth's accounts of Arthurian legends in his *History* are more similar overall in content to the accounts in the Mabinogion than they are to the accounts in *Roman de Brut*.

C Malory did not use Annales Cambriae as a source for information he presented about the *Round Table*.

D Geoffrey of Monmouth was unaware of stories of the *Round Table* when composing his *History*, though historians know that works containing such stories were available to him.

16

In Puerto Rico, it's not unusual for a city or town to be known by a nickname that corresponds to one of its notable features, like landscape, climate, famous residents, or chief exports. For example, the Puerto Rican municipality of Florida has also been called "the Town of Cayenalisa Pineapple," a nickname that alludes to what the area is well _____ the pineapple, one of its most popular exports.

Which choice completes the text so that it conforms to the conventions of Standard English?

A known: for

B known for:

C known for

D known for;

17

A region of the Soviet Union encompassing the present-day city of Tiraspol, Moldova, the Romanian-speaking Moldavian Autonomous Soviet Socialist Republic was in fact only nominally autonomous. President Mikhail Gorbachev's reforms, by helping the Moldavian Republic and other Soviet republics established along ethnolinguistic lines assert their prerogatives, _____ to the Soviet Union's breakup.

Which choice completes the text so that it conforms to the conventions of Standard English?

A which contributed

B contributed

C having contributed

D contributing

18

Minerals can be classified by how much light passes through them. For example, all (or almost all) light passes through zorite, which is classified as _____ only some light to pass through; amesite is instead classified as translucent.

Which choice completes the text so that it conforms to the conventions of Standard English?

A transparent, allowing

B transparent allowing

C transparent. Allowing

D transparent and allowing

19

The artist Jean-Honore Fragonard's painting The Progress of Love: Meeting, symbolically _____ to portray the sense of excitement that infuses new relationships, is a highly regarded example of allegory.

Which choice completes the text so that it conforms to the conventions of Standard English?

A functioning

B has functioned

C functions

D was functioning

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20

The relationship between genomes and epigenomes reveals how cells with identical DNA develop different _____ whereas the genome in each cell contains a complete DNA sequence, the epigenome consists of chemical compounds that determine which traits in the sequence will be expressed.

Which choice completes the text so that it conforms to the conventions of Standard English?

A functions:

B functions and,

C functions

D functions,

21

That the geographic center of North America lay in the state of North Dakota was conceded by all _____ establishing its precise coordinates proved more divisive.

Which choice completes the text so that it conforms to the conventions of Standard English?

A involved,

B involved;

C involved:

D involved

22

Monument 61 is the name given to a 13-ton stone statue of a human head that was created during the ancient Olmec civilization. It was discovered in Veracruz, Mexico, at the San Lorenzo archaeological site. _____ at the La Venta site in the state of Tabasco, another colossal head statue called Monument 4 was discovered.

Which choice completes the text with the most logical transition?

A As a result,

B Specifically,

C For example,

D Elsewhere,

23

In the South Asian country of Sri Lanka, the percentage of the population living in cities increased from 18.3% to 18.7% between 2015 and 2020; this upward trend in urbanization was seen throughout that part of the continent. _____ from 2015 to 2020, the overall percentage of the South Asian population living in cities rose from 34.5% to 36.6%.

Which choice completes the text with the most logical transition?

A Nevertheless,

B Indeed,

C In reality,

D Later,

24

Ugandan American professor Peter Nazareth believed that Elvis Presley's music is best understood not as a homogeneous collection but as an anthology (because Elvis showcased the contributions of a wide range of gospel, blues, and rock artists). _____ Nazareth entitled his college course on Elvis and his music, which focused on Elvis's many musical influences, "Elvis as Anthology."

Which choice completes the text with the most logical transition?

A In sum,

B To that end,

C In addition,

D That is,

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25

While researching a topic, a student has taken the following notes:

- *A copyright prevents a book's contents from being reproduced (published) without permission from the copyright holder.*
- *When a book's copyright expires, the book enters the public domain.*
- *Public domain books can be legally reproduced by anyone.*
- *The Mother's Recompense is a novel by Edith Wharton.*
- *It was published in 1925 and entered the public domain on January 1, 2021 (ninety-six years later).*

Which choice most effectively uses information from the given sentences to explain why *The Mother's Recompense* entered the public domain?

A Once *The Mother's Recompense* entered the public domain, anyone could publish its contents.

B *The Mother's Recompense*, a novel by Edith Wharton, is freely available to the public.

C Before *The Mother's Recompense* entered the public domain only its copyright holder could give permission for others to reproduce its contents.

D The copyright to *The Mother's Recompense* expired in 2021, causing the novel to enter the public domain.

26

While researching a topic, a student has taken the following notes:

- *Alfajayucan is a municipality in the state of Hidalgo, Mexico.*
- *Municipalities are governmental regions responsible for providing many public services to their residents.*
- *She completed Water Line in 2006.*
- *One service they provide is water treatment.*
- *Alfajayucan's population was 19,162 in 2020.*
- *Hidalgo is divided into 84 municipalities.*

The student wants to provide a specific example of what municipalities provide. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A Alfajayucan is one of 84 municipalities in Hidalgo, Mexico.

B In 2020, the municipality of Alfajayucan had a population of 19,162.

C The municipality Altajayucan in Hidalgo is responsible for providing many public services.

D Water treatment is just one of the many public services that municipalities provide.

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27

While researching a topic, a student has taken the following notes:

- *The Mohs scale of mineral hardness is a ten-point scale that orders minerals by hardness based on their ability to scratch other minerals.*
- *Minerals with larger numbers are harder than minerals with smaller numbers and can leave visible scratches on them.*
- *Minerals with smaller numbers are softer than minerals with larger numbers and cannot leave visible scratches on them.*
- *The mineral apatite has a Mons scale number of 5.*
- *The mineral topaz has a Mohs scale number of 8.*
- *The mineral diamond has a Mons scale number of 10.*

The student wants to compare the hardness of apatite and diamond. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. A mineral with a Mohs number of 8, like topaz, is harder than one with a Mohs number of 5, like apatite.

B. The Mohs scale of mineral hardness can be used to order topaz, apatite, and diamond by their ability to scratch other minerals.

C. Diamond can leave visible scratches on apatite, which means that diamond ds harder, than apatite.

D. Diamond can leave visible scratches on topaz, which is why diamond has a higher number than topaz on the Mohs scale of mineral hardness.

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STOP

If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.

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Math

– 22 Questions | 35 minutes –

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

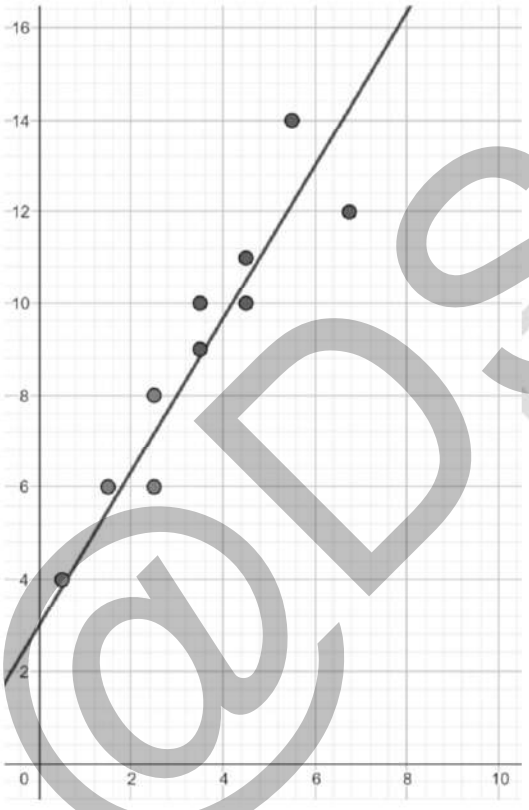
NOTES

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- *All figures lie in a plane.*
- *The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.*

1

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

A $y = \frac{5}{3}x + 3$

B $y = -2x + 1.5$

C $y = 3x + 1.5$

D $y = -3x + 1.5$

2

$3x - 4y = 5$

$x = 7$

Which is the solution (x, y) to the given systems of equations?

A $(7, 4)$

B $(7, 12)$

C $(7, -12)$

D $(7, -4)$

3

What value of x is the solution to the equation $20x + 24 = 28x$?

A $\frac{1}{4}$

B $\frac{1}{3}$

C 3

D 5

4

The function f is defined by $f(x) = \frac{7}{10}x + 74$. What is the value of $f(20)$?

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5

The function f is defined by $f(x) = 8 + \sqrt{x}$. What is the value of $f(36)$?

A 0

B 14

C 17

D 26

6

Triangles ABC and DEF are congruent, where A corresponds to D and B and E are right angles. The measure of angle A is 27°. What is the measure, in degrees, of angle F?

7

At a convention center, there are a total of 325 visitors. Each visitor is located in either room A, room B or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.48, and the probability of selecting a visitor who is located in room B is 0.16. How many visitors are located in room C?

A 25

B 36

C 104

D 117

8

$g(m) = -0.05m + 14.1$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

A 0.05

B 14.1

C 20

D 282

9

What is the center of the circle in the xy-plane defined by the equation $(x-9)^2 + (y+6)^2 = 81$?

A $(-9,-6)$

B $(-9,6)$

C $(9,-6)$

D $(9,6)$

10

If $2x + 3 = 9$, what is the value of $4x - 4$?

11

$f(x) = 6,000(0.7)^x$

A conservation scientist implemented a program to reduce a certain invasive insect population in an area. The given function estimates this insect species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of $6,000$ in this context?

A The estimated percent decrease in the insect population for this species and area every 8 years after 2008

B The estimated percent decrease in the insect population for this species and area each year after 2008

C The estimated insect population for this species and area 8 years after 2008

D The estimated initial insect population for this species and area in 2008

12

Line r is defined by the equation $4x - 5y = 8$. Line s is parallel to line r in the xy-plane. What is the slope of line s?

A $\frac{5}{4}$

B $\frac{4}{5}$

C -4

D -5

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13

The cost to rent a bus from Company X is $\$750$ for the first 2 hours and an additional $\$50$ per hour for each hour after the first 2 hours. If the total cost to rent the bus from Company X for t hours, where $t > 2$, is $\$1,050$, which equation represents this situation?

A $750(t-2)+50t=1,050$

B $750(2t)+50t=1,050$

C $750+50(t-2)=1,050$

D $750+50(2t)=1,050$

14

$$y=-\frac{1}{4}x^2+x+28$$

The given equation models a company's active projects over 6 months, where y is the estimated number of active projects x months after the end of April 2012, where $0<x<6$. Which statement is the best interpretation of the y-intercept of the graph of this equation in the xy-plane?

A At the end of April 2012, the estimated number of active projects was 0.

B At the end of April 2012, the estimated number of active projects was 28.

C At the end of May 2012, the estimated number of active projects was 0.

D At the end of May 2012, the estimated number of active projects was 28.

15

In right triangle RST, the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{4\sqrt{2}}{9}$. What is the value of $\cos(S)$?

A $\frac{4\sqrt{2}}{9}$

B $\frac{4\sqrt{2}}{7}$

C $\frac{7\sqrt{2}}{8}$

D $\frac{9\sqrt{2}}{8}$

16

$19.5x+29.75y=394$

Odalys ordered mulch and river rock, which cost a total of $\$394$, for her home. The given equation represents the relationship between the number of cubic yards of mulch, x, and the number of tons of river rock, y Odalys ordered. How much more, in dollars, did a ton of river rock cost Odalys than a cubic yard of mulch? Enter decimal form only.

17

A hemisphere is half of a sphere. If a hemisphere has a radius of 50 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

A $5,200$

B $20,900$

C $196,300$

D $261,800$

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18

In triangle ABC, angle B is a right angle, and the length of BC is 248 millimeters. If $\cos(A) = \frac{3}{5}$, what is the length, in millimeters, of AB?

A 62

B 186

C 248

D 310

19

In a certain state, the population of pheasants, a type of bird, is estimated each year by counting the number of pheasants observed along certain roads in the state. On average, each year from 2005 to 2015 the number of pheasants counted per mile of road decreased by 3.5 of the number of pheasants per mile of road the previous year. Based on this average, if there were 6.97 pheasants per mile of road in this state in 2005, which of the following best approximates the number of pheasants per mile of road in 2015?

A $0.965(6.97)^{10}$

B $1.035(6.97)^{10}$

C $6.97(0.035)^{10}$

D $6.97(0.965)^{10}$

20

To study fluctuations in leaf water potential, samples of wood were taken from 25 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 2.00 centimeters. If this cube has a mass of 2.72 grams, what is the density of this cube, in grams per cubic centimeter? Enter decimal form only.

21

x	y
-11	-25
9	55

The table shows two values of x and their corresponding values of y. The graph of the linear equation representing this relationship passes through the point $\left(\frac{1}{3}, a\right)$. What is the value of a? Enter decimal form only and round to two decimal places.

22

Which system of linear equations has no solution?

A
$-2x + 8y = -5$
$2x - 8y = 5$

B
$2x - 8y = 5$
$3x + 9y = 6$

C
$2x - 8y = 5$
$-6x + 24y = -15$

D
$-2x + 8y = 5$
$4x - 16y = 10$

STOP

If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.

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Math

– 22 Questions | 35 minutes –

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

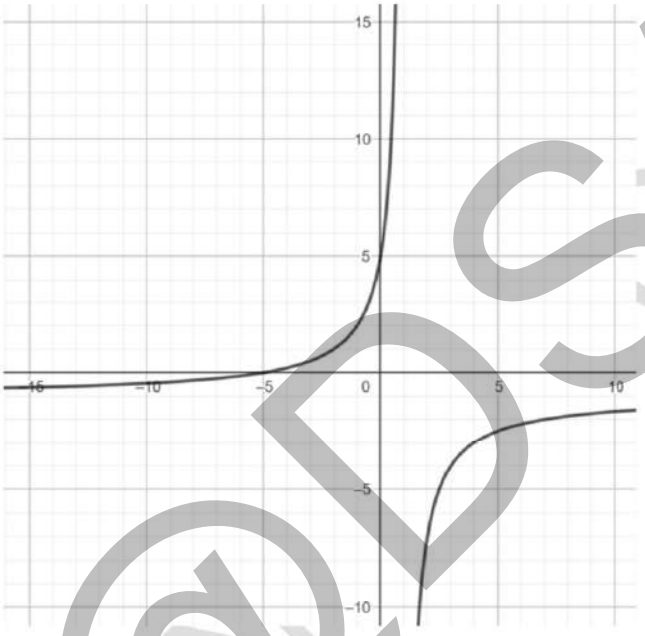
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1

The graph of $y = f(x)$ is shown in the xy-plane. What is the value of $f(0)$?



2

Which expression is equivalent to $(9nr^2 + 3nr) + (6n^2r + 5nr)$?

A $15n^2r^2 + 8nr$

B $15n^3r^3 + 8n^2r^2$

C $54n^3r^3 + 15n^2r^2$

D $9nr^2 + 6n^2r + 8nr$

3

If $\frac{2a}{b} = 5.8$ and $\frac{a}{bn} = 23.2$, what is the value of $\frac{1}{n}$?

4

$$v = -\frac{w}{161x}$$

The given equation relates the distinct positive numbers v, w and x. Which equation correctly expresses w in terms of v and x?

A $w = -161vx$

B $w = -\frac{161v}{x}$

C $w = -\frac{x}{161v}$

D $w = v + 161x$

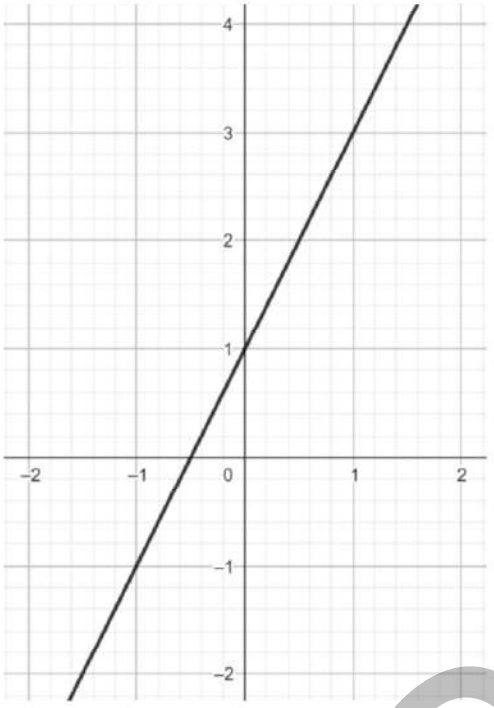
5

A real estate company offers a series of three webinars. 1,250 people attended the first webinar. 68 of the people who attended the first webinar attended the second webinar, and 32 of the people who attended the first and second webinars attended the third webinar. How many people attended all three webinars?

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6

The graph of $y = f(x)$ is shown. What is the y-intercept of the graph of $y = f(x) + 13$?



A $(0, -12)$

B $(0, 14)$

C $(1, 13)$

D $(1, 14)$

7

The function f is defined by $f(x) = |x - 6x|$. What value of a satisfies $f(7) - f(a) = -15$?

A -22

B 3

C 10

D 75

8

If $-9(15 - 3x) + 5 = -10(15 - 3x) + 21$, what is the value of $15 - 3x$?

9

The minimum value of x is 11 less than 8 times another number n. Which inequality shows the possible values of x?

A $x \leq 8n - 11$

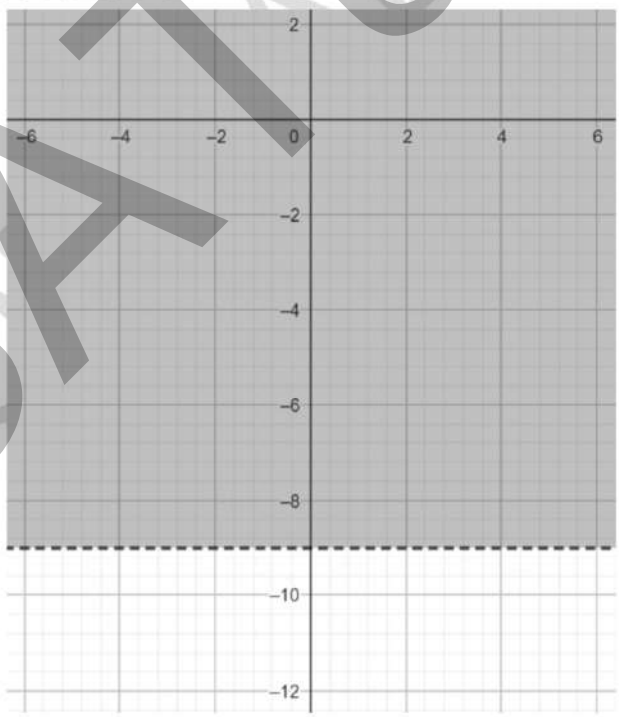
B $x \geq 8n - 11$

C $x \leq 11 - 8n$

D $x \geq 11 - 8n$

10

The shaded region shown represents the solutions to the inequality $-18y < c$, where c is a constant. What is the value of c?



A 162

B 9

C -9

D -162

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11

At a constant temperature, a scuba diver combines two different concentrations of nitrox mix (oxygen-nitrogen gas mixture) to fill a tank to a total pressure of 130 atmospheres (*atm*), of which 34 is contributed by oxygen. Nitrox mix *A* contributes x *atm* of pressure to the tank, of which 22 is contributed by oxygen. Nitrox mix *B* contributes y *atm* of pressure to the tank, of which 48 is contributed by oxygen. Which system of equations represents this situation?

A $x + y = 130$
$0.48x + 0.22y = 0.34(130)$

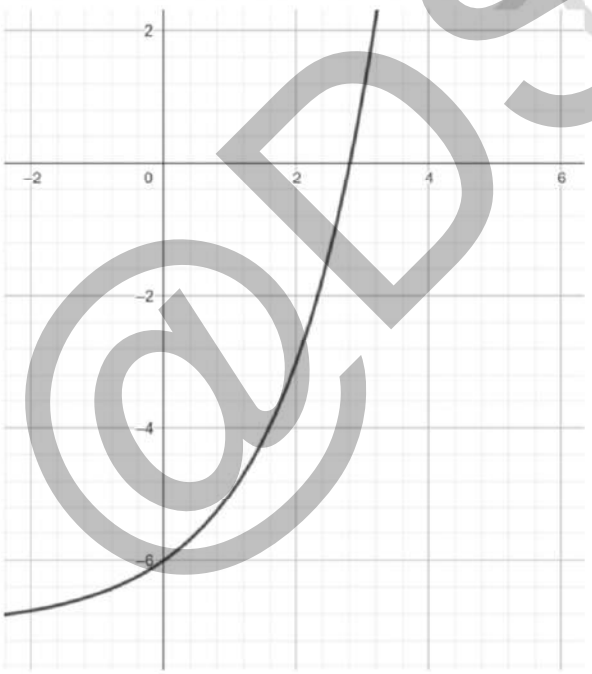
B $x + y = 130$
$0.22x + 0.48y = 0.34(130)$

C $x + y = 0.34(130)$
$0.48x + 0.22y = 130$

D $x + y = 0.34(130)$
$0.22x + 0.48y = 130$

12

The graph of the equation $y = 2^x + k$ is shown, where k is a constant. What is the value of k?



A -7

B -6

C -5

D -4

13

A certain town has an area of 3.99 square miles. What is the area, in square yards, of this town (1 mile = 1,760 yards)?

A 441

B 7,022

C 776,341

D 12,359,424

14

The function f is defined by $f(x) = 4x^2$. What is the value of $f(11)$?

A 44

B 52

C 88

D 484

15

The function f gives the product of a number, x, and a number that is 66 more than x. Which equation defines f?

A $f(x) = x^2 + x + 66$

B $f(x) = x^2 + 66$

C $f(x) = x^2 + 66x$

D $f(x) = x^2 + 66x + 66$

16

One of the factors of $4x^3 + 88x^2 + 468x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b?

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17

Alex engages in up to 3 types of exercise each week for a total of 9 hours while training for a triathlon. Alex swims for the same number of minutes each week. The equation $y = 540 - x - 200$ represents the situation where Alex bikes for x minutes during a week and runs for any remaining training time y, in minutes. If this equation is graphed in the xy-plane, which statement is the best interpretation of the x-intercept of the graph?

A. During a week when Alex swims for 340 minutes, he bikes for 200 minutes.

B. Each week, Alex bikes and runs for a total of 340 minutes.

C. During a week when Alex doesn't bike, he runs for 340 minutes.

D. During a week when Alex doesn't run, he bikes for 340 minutes.

18

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 190)^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

A. $(-18x + 380)^\circ$

B. $(-18x + 740)^\circ$

C. $(18x - 380)^\circ$

D. 180°

19

$$\frac{4}{7}(4x+7)\left(x+\sqrt{4k+7}\right)\left(x-\sqrt{4k+7}\right)=0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 77. What is the value of k? Enter decimal form only.

20

At the time of posting a video, a social media channel had 100 subscribers. Each day for five days after the video was posted, the number of subscribers tripled from the number the previous day. Which equation gives the total number of subscribers, n, to the channel d days after the video was posted?

A. $n = (100)^d$

B. $n = 100(3)^d$

C. $n = 100\left(\frac{1}{3}\right)^d$

D. $n = (100)^3 + d$

21

	Volume *(cubic units)*
Right circular cylinder A	112π
Right circular cylinder B	$3,024\pi$

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 4 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.

22

Function f is a quadratic function. The graph of $y = f(x)$ in the xy-plane has a vertex at $(-8,1)$, contains the point $(-9,-1)$, and has a y-intercept at $(0,a)$. The graph of $y = 6f(x)$ has a y-intercept at $(0,b)$. What is the positive difference between a and b?

STOP

If you finish before time is called, you may check your work on this module only.

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X

March 2024 DSAT

Do not turn to any other module in the test.

Digital SAT Reference Test

March 2024 *Question Booklet*

Test date: ..... / ..... / .....

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SCOREBOARD

Reading and Writing *Math*

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Module 1	/ 27	/ 22
Module 2	/ 27	/ 22
Total points		/ 98

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1	C
2	D
3	D
4	B
5	D
6	D
7	B
8	D
9	C
10	A
11	A
12	A
13	D
14	D
15	D
16	A
17	C
18	A
19	C
20	B
21	A
22	D
23	B
24	A
25	B
26	A
27	D

rw1

READING AND WRITING

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MARCH SAT...

Chưa được lưu

MODULE 1

MODULE 2

1	D
2	A
3	B
4	A
5	B
6	A
7	A
8	B
9	B
10	D
11	A
12	C
13	D
14	C
15	C
16	B
17	B
18	C
19	A
20	A
21	B
22	D
23	B
24	B
25	D
26	D
27	C

RW 2

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1	5
2	D
3	8
4	A
5	272
6	B
7	C
8	16
9	B
10	A
11	B
12	A
13	D
14	D
15	C
16	9
17	D
18	A
19	9.25
20	B
21	704
22	635
1	A
2	A
3	C
4	88
5	B
6	63
7	D
8	A
9	C
10	8
11	D

m1

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Chưa được lưu

14	D
15	C
16	9
17	D
18	A
19	9.25
20	B
21	704
22	635
1	A
2	A
3	C
4	88
5	B
6	63
7	D
8	A
9	C
10	8
11	D
12	B
13	C
14	B
15	A
16	10.25
17	D
18	B
19	D
20	0.34
21	20.33
22	D

m2

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